

Name: _____

A Level Mathematics : Pure Mathematics

Topic 4 :Sequences and series - Sequences and series in modelling

Date:

Time:

Total marks available:

Total marks achieved: _____

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Questions

Q1.

The value, £ V , of a vintage car t years after it was first valued on 1st January 2001, is modelled by the equation

$$V = Ap^t \quad \text{where } A \text{ and } p \text{ are constants}$$

Given that the value of the car was £32 000 on 1st January 2005 and £50 000 on 1st January 2012

(a) (i) find p to 4 decimal places,

(ii) show that A is approximately 24 800

(4)

(b) With reference to the model, interpret

(i) the value of the constant A ,

(ii) the value of the constant p .

(2)

Using the model,

(c) find the year during which the value of the car first exceeds £100 000

(4)

(Total for question = 10 marks)

Q2.

A competitor is running a 20 kilometre race.

She runs each of the first 4 kilometres at a steady pace of 6 minutes per kilometre. After the first 4 kilometres, she begins to slow down.

In order to estimate her finishing time, the time that she will take to complete each subsequent kilometre is modelled to be 5% greater than the time that she took to complete the previous kilometre.

Using the model,

(a) show that her time to run the first 6 kilometres is estimated to be 36 minutes 55 seconds,

(2)

(b) show that her estimated time, in minutes, to run the r th kilometre, for $5 \leq r \leq 20$, is

$$6 \times 1.05^{r-4}$$

(1)

(c) estimate the total time, in minutes and seconds, that she will take to complete the race.

(4)

(Total for question = 7 marks)

Q3.

A car has six forward gears.

The fastest speed of the car

- in 1st gear is 28 km h⁻¹
- in 6th gear is 115 km h⁻¹

Given that the fastest speed of the car in successive gears is modelled by an **arithmetic sequence**,

(a) find the fastest speed of the car in 3rd gear.

(3)

Given that the fastest speed of the car in successive gears is modelled by a **geometric sequence**,

(b) find the fastest speed of the car in 5th gear.

(3)

(Total for question = 6 marks)

Q4.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A geometric series has common ratio r and first term a .

Given $r \neq 1$ and $a \neq 0$

(a) prove that

$$S_n = \frac{a(1 - r^n)}{1 - r} \quad (4)$$

Given also that S_{10} is four times S_5

(b) find the exact value of r .

(4)

(Total for question = 8 marks)

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